

1. Matrix Notation.

- (a) Show that $A^*A = I$ if and only if the columns of A are orthonormal.
- (b) Show that a vector $\mathbf{x}$ is orthogonal to every column of A if and only if $A^*\mathbf{x} = \mathbf{0}$.
- (c) Suppose that $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ for some vectors $\mathbf{v}_i$ and scalars λ_i . Show how to express this with a single matrix equation $AV = V\Lambda$.
- (d) Suppose that $A = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ for some vectors $\mathbf{u}_i, \mathbf{v}_i$ and scalars σ_i . Show how to express this with a single matrix equation $A = U\Sigma V^*$.

(a): The ij entry of the identity matrix I is the Kronecker delta:

$$(ij \text{ entry of } I) = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Recall that for any column vectors $\mathbf{u}, \mathbf{v}$ of the same length we have $\mathbf{u}^*\mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle$. Let $\mathbf{a}_i$ be the column vectors of the matrix A , so that $\mathbf{a}_i^*$ are the rows of A^* . Then we have

$$(ij \text{ entry of } A^*A) = (i\text{-th row of } A^*)(j\text{-th column of } A) = \langle \mathbf{a}_i, \mathbf{a}_j \rangle.$$

But two matrices are equal if and only if their entries are equal, hence

$$A^*A = I \iff \langle \mathbf{a}_i, \mathbf{a}_j \rangle = \delta_{ij}.$$

(b): Let $\mathbf{a}_i$ be the column vectors of A , so that $\mathbf{a}_i^*$ are the row vectors of A^* . The i th entry of the vector $A^*\mathbf{x}$ is $\mathbf{a}_i^*\mathbf{x}$, which equals $\langle \mathbf{a}_i, \mathbf{x} \rangle$. Hence

$$A^*\mathbf{x} = \mathbf{0} \iff \langle \mathbf{a}_i, \mathbf{x} \rangle = 0 \text{ for all } i.$$

(c): Let $\mathbf{v}_j$ be the j -th column vector of V , and let Λ be the square matrix with ii entry equal to λ_i and all other entries equal to zero. The j -th column vector of AV is $A\mathbf{v}_j$. The j -th column vector of Λ is $\lambda_j\mathbf{e}_j$, hence the j -th column vector of $V\Lambda$ is $V(\lambda_j\mathbf{e}_j) = \lambda_j V\mathbf{e}_j = \lambda_j\mathbf{v}_j$. But two matrices are equal if and only if they have the same column vectors, hence

$$AV = V\Lambda \iff A\mathbf{v}_j = \lambda_j\mathbf{v}_j \text{ for all } j.$$

(d): Let $\mathbf{u}_i$ be the column vectors of U , let $\mathbf{v}_i$ be the column vectors of V (so $\mathbf{v}_i^*$ are the row vectors of V^*) and let Σ be the matrix whose ii entry is σ_i and all other entries are zero. Note that the i -th column vector of $U\Sigma$ is $\sigma_i\mathbf{u}_i$. If $A = U\Sigma V^*$ then we have

$$A = \sum_i (i\text{-th column of } U\Sigma)(i\text{-th row of } V^*) = \sum_i (\sigma_i\mathbf{u}_i)\mathbf{v}_i^*.$$

2. Projection. Let $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ with $\langle \mathbf{v}, \mathbf{u} \rangle = \mathbf{v}^*\mathbf{u} \neq 0$ and consider the $n \times n$ matrix

$$P = \frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}\mathbf{v}^*.$$

- (a) Show that $P^2 = P$. [Hint: Use associativity of matrix multiplication.]
- (b) Show that $P(t\mathbf{u}) = t\mathbf{u}$ for all scalars $t \in \mathbb{C}$.
- (c) If $\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^*\mathbf{w} = 0$, show that $P\mathbf{w} = \mathbf{0}$.
- (d) If $\mathbf{v} = \mathbf{u}$, show that the linear function $\mathbb{C}^n \rightarrow \mathbb{C}^n$ defined by $\mathbf{x} \mapsto P\mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto the subspace (complex line) $\{t\mathbf{u} : t \in \mathbb{C}\}$.

(a): We use the fact that $\mathbf{v}^*\mathbf{u}$ is a scalar (a 1×1 matrix), so it commutes with all matrices:

$$P^2 = \left(\frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}\mathbf{v}^* \right) \left(\frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}\mathbf{v}^* \right) = \frac{1}{(\mathbf{v}^*\mathbf{u})^2} \mathbf{u}(\mathbf{v}^*\mathbf{u})\mathbf{v}^* = \frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}\mathbf{v}^* = P.$$

(b): Again, we use the fact that $\mathbf{v}^*\mathbf{u}$ is a scalar:

$$P(t\mathbf{u}) = \frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}\mathbf{v}^*(t\mathbf{u}) = t \frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}(\mathbf{v}^*\mathbf{u}) = t\mathbf{u}.$$

(c): If $\mathbf{v}^*\mathbf{w} = 0$ then

$$P\mathbf{w} = \frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}(\mathbf{v}^*\mathbf{w}) = \frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}(0) = \mathbf{0}.$$

(d): In the previous section, we defined the projection of $\mathbf{x}$ onto the line spanned by $\mathbf{u}$ by

$$\frac{\langle \mathbf{x}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u}.$$

Observe that this is the same as the matrix product¹

$$\left(\frac{1}{\mathbf{u}^*\mathbf{u}} \mathbf{u}\mathbf{u}^* \right) \mathbf{x} = \frac{1}{\mathbf{u}^*\mathbf{u}} \mathbf{u}(\mathbf{u}^*\mathbf{x}) = \frac{\mathbf{u}^*\mathbf{x}}{\mathbf{u}^*\mathbf{u}} \mathbf{u}.$$

Remark: More generally, we can view the matrix $\frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}\mathbf{v}^*$ as *non-orthogonal projection*. It projects every point onto the line $t\mathbf{u}$ in a direction that is orthogonal to $\mathbf{v}$.

3. Special Matrices. For any real number t , consider the following three real 2×2 matrices:

$$R_t = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \quad F_t = \begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}, \quad P_t = \begin{pmatrix} \cos^2 t & \cos t \sin t \\ \cos t \sin t & \sin^2 t \end{pmatrix}.$$

(a) Describe what each matrix does geometrically. [Hint: Rotate, reFlect, Project.]

(b) Use trig identities to prove the following matrix identities:

$$F_t^2 = I, \quad P_t^2 = P_t, \quad F_{2t} + I = 2P_t, \quad R_s R_t = R_{s+t}, \quad F_s F_t = R_{s-t}.$$

(c) Give geometric explanations for these identities.

See the course notes for solutions.

4. Unitary and Orthogonal Matrices. A complex square matrix $U \in \mathbb{C}^{n \times n}$ is called *unitary* if $U^*U = I$. Unitary matrices with real entries are called *orthogonal* matrices. [In this case we have $U^T U = I$ because $U^* = U^T$ for real matrices.]

(a) Show that a product of unitary matrices is unitary.

(b) Show that unitary matrices preserve distance. That is, prove that we have $\|U\mathbf{x} - U\mathbf{y}\| = \|\mathbf{x} - \mathbf{y}\|$ for all unitary matrices $U \in \mathbb{C}^{n \times n}$ and vectors $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$.

(c) Show that the matrices R_t and F_t from the previous problem are orthogonal.

(d) Show that any 2×2 orthogonal matrix is equal to R_t or F_t for some t . [Hint: Let $U \in \mathbb{R}^{2 \times 2}$ have columns $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^2$. If $U^T U = I$ then $\mathbf{u}_1, \mathbf{u}_2$ is an orthonormal set. If $\|\mathbf{u}_1\| = 1$ then we can write $\mathbf{u}_1 = (\cos t, \sin t)$ for some unique angle $0 \leq t < 2\pi$.]

¹I see that our previous formula needs to be modified for the complex case. It should be $\frac{\langle \mathbf{u}, \mathbf{x} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u}$.

(a): If $U^*U = I$ and $V^*V = I$, and if the matrix product UV is defined, then we have

$$(UV)^*(UV) = V^*U^*UV = V^*IV = V^*V = I.$$

Here we used the associative property and the fact that $(UV)^* = V^*U^*$.

(b): If $U^*U = I$ then for all vectors $\mathbf{z}$ we have

$$\|U\mathbf{z}\|^2 = (U\mathbf{z})^*(U\mathbf{z}) = \mathbf{z}^*U^*U\mathbf{z} = \mathbf{z}^*I\mathbf{z} = \mathbf{z}^*\mathbf{z} = \|\mathbf{z}\|^2,$$

and hence $\|U\mathbf{z}\| = \|\mathbf{z}\|$. Then substitute $\mathbf{z} = \mathbf{x} - \mathbf{y}$ to get

$$\|\mathbf{x} - \mathbf{y}\| = \|U(\mathbf{x} - \mathbf{y})\| = \|U\mathbf{x} - U\mathbf{y}\|.$$

5. Frobenius Norm. For any complex matrix $A = (a_{ij})$ we define the Frobenius norm by

$$\|A\|_F := \sqrt{\sum_{i,j} |a_{ij}|^2}.$$

(a) Let $\|\cdot\|$ denote the standard Hermitian norm on complex vectors. If $\mathbf{a}_1, \dots, \mathbf{a}_m$ are the columns and $\mathbf{r}_1^*, \dots, \mathbf{r}_\ell^*$ are the rows of a matrix $A \in \mathbb{C}^{\ell \times m}$, show that

$$\|A\|_F^2 = \|\mathbf{a}_1\|^2 + \dots + \|\mathbf{a}_m\|^2 = \|\mathbf{r}_1\|^2 + \dots + \|\mathbf{r}_\ell\|^2.$$

(b) For all matrices $A \in \mathbb{C}^{\ell \times m}$ and vectors $\mathbf{x} \in \mathbb{C}^m$, show that

$$\|A\mathbf{x}\| \leq \|A\|_F \|\mathbf{x}\|.$$

[Hint: If $\mathbf{r}_i^*$ is the i th row of A , then $\mathbf{r}_i^* \mathbf{x}$ is the i th entry of $A\mathbf{x}$. From Cauchy-Schwarz we know that $|\mathbf{r}_i^* \mathbf{x}|^2 \leq \|\mathbf{r}_i\|^2 \|\mathbf{x}\|^2$. Now use the row formula from part (a).]

(c) For any matrices $A \in \mathbb{C}^{\ell \times m}$ and $B \in \mathbb{C}^{m \times n}$, combine parts (a) and (b) to prove that

$$\|AB\|_F \leq \|A\|_F \|B\|_F.$$

[Hint: Let $\mathbf{b}_j \in \mathbb{C}^m$ denote the j th column of B , so that $A\mathbf{b}_j$ is the j th column of AB . Now use part (b) and the column formula from part (a).]

(a): Suppose $A \in \mathbb{C}^{m \times n}$ and suppose the j -th column vector $\mathbf{a}_j$, which has coordinates $a_{1j}, \dots, a_{mj}$, so that

$$\|\mathbf{a}_j\|^2 = \mathbf{a}_j^* \mathbf{a}_j = a_{1j}^* a_{1j} + \dots + a_{mj}^* a_{mj} = |a_{1j}|^2 + \dots + |a_{mj}|^2 = \sum_i |a_{ij}|^2.$$

Then grouping the double sum by columns gives

$$\|A\|_F^2 = \sum_{i,j} |a_{ij}|^2 = \sum_j \sum_i |a_{ij}|^2 = \sum_j \|\mathbf{a}_j\|^2.$$

For the other identity, note that the row vector $\mathbf{r}_i^*$ has entries $a_{i1}, \dots, a_{in}$, so the column vector $\mathbf{r}_i$ has entries $a_{i1}^*, \dots, a_{in}^*$, hence

$$\|\mathbf{r}_i\|^2 = a_{i1} a_{i1}^* + \dots + a_{in} a_{in}^* = |a_{i1}|^2 + \dots + |a_{in}|^2 = \sum_j |a_{ij}|^2.$$

Then

$$\|A\|_F^2 = \sum_{i,j} |a_{ij}|^2 = \sum_i \sum_j |a_{ij}|^2 = \sum_i \|\mathbf{r}_i\|^2.$$

(b): The Cauchy-Schwarz identity says that $|\mathbf{r}_i^* \mathbf{x}| \leq \|\mathbf{r}_i\| \|\mathbf{x}\|$ for all i , hence

$$\|A\mathbf{x}\|^2 = \sum_i |\mathbf{r}_i^* \mathbf{x}|^2 \leq \sum_i \|\mathbf{r}_i\|^2 \|\mathbf{x}\|^2 = \left(\sum_i \|\mathbf{r}_i\|^2 \right) \|\mathbf{x}\|^2 = \|A\|_F^2 \|\mathbf{x}\|^2.$$

Now take the square root of both sides. Note that we used the row identity from part (a).

(c): Let $\mathbf{b}_j$ be the columns of B then $A\mathbf{b}_j$, so that $A\mathbf{b}_j$ are the columns of AB . From (b) we have $\|A\mathbf{b}_j\| \leq \|A\|_F \|\mathbf{b}_j\|$ for all j . Then the column identity from part (a) implies that

$$\|AB\|_F^2 = \sum_j \|A\mathbf{b}_j\|^2 \leq \sum_j \|A\|_F^2 \|\mathbf{b}_j\|^2 = \|A\|_F^2 \left(\sum_j \|\mathbf{b}_j\|^2 \right) = \|A\|_F^2 \|B\|_F^2.$$

Finally, take the square root of both sides.

Challenge Problem (Geometric Series of Matrices). Let A be a square matrix with $\|A\|_F < 1$. In this problem you will show that $I - A$ is invertible, with a power series expansion that converges with respect to the Frobenius norm:²

$$(I - A)^{-1} = I + A + A^2 + \cdots = \sum_{k \geq 0} A^k.$$

- (a) Show that $\|A^n\|_F \leq \|A\|_F^n$. Use this to show that A^n converges to the zero matrix
- (b) Let $S_n = \sum_{k=0}^n A^k$. If $m < n$, show that $\|S_n - S_m\|_F \leq \sum_{k=m+1}^n \|A\|_F^k$. From the convergence of the scalar geometric series, this proves that S_n is a Cauchy sequence, hence it converges to some matrix T .
- (c) Observe that $S_n(I - A) = (I - A)S_n = I - A^{n+1}$. Use (a) to show that the right side converges to I and use (b) to show that $S_n(I - A)$ and $(I - A)S_n$ converge to $T(I - A)$ and $(I - A)T$, respectively. Hence $T(I - A) = (I - A)T = I$, as desired.
- (d) **Application: Absorbing Markov Chains.** Consider a partitioned matrix

$$M = \left(\begin{array}{c|c} I & 0 \\ \hline R & Q \end{array} \right),$$

where I is an identity matrix, R is any rectangular matrix and Q is a square matrix satisfying $\|Q\|_F < 1$. Use the geometric series to show that

$$M^n \rightarrow \left(\begin{array}{c|c} I & 0 \\ \hline (I - Q)^{-1}R & 0 \end{array} \right) \quad \text{as } n \rightarrow \infty.$$

[Hint: Compute the first few powers of M and observe a pattern.]

- (e) **Example: Gambler's Ruin.** Consider a game where at any stage you have d dollars for some $d \in \{1, 2, 3, 4\}$. The game consists of repeatedly flipping a coin that lands heads with probability p and tails with probability $q = 1 - p$. Each time you get heads you gain a dollar and each time you get tails you lose a dollar. The game ends when $d = 1$ or $d = 4$. Consider the matrix $M = (m_{ij})$, where m_{ij} is the probability that $d = j$ next turn, assuming that $d = i$ this turn, and the matrix $P = (p_{ij})$, where p_{ij} is the probability that the game ultimately ends with $d = j$, assuming that it started with $d = i$. It follows from basic rules of probability that

$$P = \lim_{n \rightarrow \infty} M^n.$$

Use part (d) to compute the entries of P . [Hint: Reorder the states as 1, 4, 2, 3.]

²To be precise, for any sequence of matrices $X_1, X_2, \dots$ and for any matrix Y , we say that X_n converges to Y if and only if $\|X_n - Y\|_F$ converges to the number zero. It follows from the completeness of the complex numbers that if $\|X_n - X_m\|_F$ gets arbitrarily small for n and m arbitrarily large (i.e., if X_n is a *Cauchy sequence*) then there exists some matrix Y such that $X_n \rightarrow Y$.

We will prove the results for a general sub-multiplicative matrix norm $\| \cdot \|$, not necessarily the Frobenius norm.

(a): If $\|A^k\| \leq \|A\|^k$, then we have

$$\|A^{k+1}\| = \|AA^k\| \leq \|A\|\|A^k\| \leq \|A\|\|A\|^k = \|A\|^{k+1}.$$

It follows by induction that $\|A^n\| \leq \|A\|^n$ for all $n \geq 1$.³ If $\|A\| < 1$ then this implies that

$$\|A^n - O\| = \|A^n\| \leq \|A\|^n \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

In other words, $A_n \rightarrow O$.

(b): Let $S_n = I + A + A^2 + \cdots + A^n$. If $1 \leq m < n$ then

$$\begin{aligned} S_n - S_m &= (I + A + A^2 + \cdots + A^n) - (I + A + A^2 + \cdots + A^m) \\ &= A^{m+1} + A^{m+2} + \cdots + A^n. \end{aligned}$$

Suppose $r := \|A\|$. Then from part (a) and the triangle inequality we have

$$\|S_n - S_m\| \leq \|A\|^{m+1} + \cdots + \|A\|^n = r^{m+1} + \cdots + r^n = r^{m+1} \cdot \frac{1 - r^{n-m}}{1 - r}.$$

If $r < 1$ then we can make the right hand side arbitrarily small by taking m sufficiently large. Hence S_n is a Cauchy sequence. Since the complex numbers are complete, Cauchy sequences converge, hence there exists a matrix T such that $\|S_n - T\| \rightarrow 0$ as $n \rightarrow \infty$.

(c): Note that

$$S_n(I - A) = (I + A + A^2 + \cdots + A^n) - (A + A^2 + \cdots + A^{n+1}) = I - A^{n+1}.$$

Thus for any $n \geq 1$ we have

$$\begin{aligned} \|I - T(I - A)\| &= \|I - S_n(I - A) + S_n(I - A) - T(I - A)\| \\ &= \|A^{n+1} + (S_n - T)(I - A)\| \\ &\leq \|A^{n+1}\| + \|(S_n - T)(I - A)\| \\ &\leq \|A\|^{n+1} + \|S_n - T\|\|I - A\|. \end{aligned}$$

Since $\|A\|^{n+1} \rightarrow 0$ and $\|S_n - T\| \rightarrow 0$ as $n \rightarrow \infty$, the right hand side goes to zero. Then from the squeeze theorem we have $\|I - T(I - A)\| = 0$, hence $T(I - A) = I$. The proof of $(I - A)T = I$ is similar.⁴ We have shown that T is a two-sided inverse of $I - A$.

(d): We will show by induction that

$$M^n = \left(\begin{array}{c|c} I & 0 \\ \hline (I + Q + \cdots + Q^{n-1})R & Q^n \end{array} \right).$$

³I should have mentioned in the problem that $n \neq 0$ because we usually interpret A^0 as the identity matrix, and $\|I\| \leq 1$ is not generally true. For example, if I_k is the $k \times k$ identity matrix then the Frobenius norm is $\|I_k\|_F = \sqrt{k}$.

⁴Later we will prove that $AB = I$ implies $BA = I$ when A and B are square.

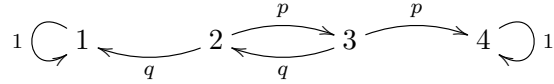
Indeed, this statement holds for $n = 1$, and it holds for a given n , then it holds for $n + 1$:

$$\begin{aligned}
M^{n+1} &= MM^n \\
&= \left(\begin{array}{c|c} I & 0 \\ \hline R & Q \end{array} \right) \left(\begin{array}{c|c} I & 0 \\ \hline (I+Q+\dots+Q^{n-1})R & Q^n \end{array} \right) \\
&= \left(\begin{array}{c|c} I & 0 \\ \hline R+Q(I+Q+\dots+Q^{n-1})R & QQ^n \end{array} \right) \\
&= \left(\begin{array}{c|c} I & 0 \\ \hline (I+Q+\dots+Q^n)R & Q^{n+1} \end{array} \right).
\end{aligned}$$

If $\|Q\| < 1$ then from the previous results, we have $Q^n \rightarrow O$ and $(I+Q+\dots+Q^{n-1})R \rightarrow (I-Q)^{-1}R$. Since block matrices converge blockwise,⁵ this implies that

$$M^n \rightarrow \left(\begin{array}{c|c} I & 0 \\ \hline (I-Q)^{-1}R & 0 \end{array} \right).$$

(e): The game can be drawn schematically as follows, where the vertices are the possible states and the labels on the arrows represent transition probabilities:



The transition probabilities can be encoded by the matrix

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ q & 0 & p & 0 \\ 0 & q & 0 & p \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This is called a “stochastic matrix”, or a “Markov chain”. The key property is that the entries in each row sum to 1 because they are probabilities. It follows from basic facts of probability that the ij entry of the n th power M^n equals the probability of being in state j exactly n steps from now, assume you are currently in state j . Thus we want to compute the limit of M^n as $n \rightarrow \infty$. In order to apply part (d) we need to reorder the states so that the “absorbing states” come first. If we order the states as 1, 4, 2, 3 then the matrix becomes

$$M = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline q & 0 & 0 & p \\ 0 & p & q & 0 \end{array} \right),$$

so that

$$Q = \begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix} \quad \text{and} \quad R = \begin{pmatrix} q & 0 \\ 0 & p \end{pmatrix}.$$

Since $0 < p < 1$ and $p + q = 1$ we have $\|Q\|_F = \sqrt{1/p^2 + 1/q^2} < 1$ with respect to the Frobenius norm, so the result in part (d) applies. Then we compute

$$(I - Q)^{-1} = \begin{pmatrix} 1 & -p \\ -q & 1 \end{pmatrix}^{-1} = \frac{1}{1 - pq} \begin{pmatrix} 1 & p \\ q & 1 \end{pmatrix}$$

⁵Actually, this needs to be proved. It is fairly straightforward for the Frobenius norm. And it holds for other norms because any two norms on a finite dimensional vector space are “topologically equivalent”, i.e., they lead to the same notions of continuity and convergence.

and

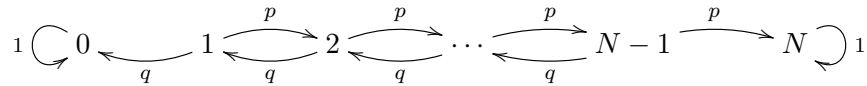
$$(I - Q)^{-1}R = \frac{1}{1 - pq} \begin{pmatrix} 1 & p \\ q & 1 \end{pmatrix} \begin{pmatrix} q & 0 \\ 0 & p \end{pmatrix} = \frac{1}{1 - pq} \begin{pmatrix} q & p^2 \\ q^2 & p \end{pmatrix}.$$

It follows that

$$M^n \rightarrow \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline q/(1-pq) & p^2/(1-pq) & 0 & 0 \\ q^2/(1-pq) & p/(1-pq) & 0 & 0 \end{array} \right).$$

Note that the entries in each row of the limit matrix still sum to 1. The entries give the long term probabilities of ending up in a certain state. For example, if you begin in state 2 then the probability that you eventually end up in the “absorbing state” 4 is $p^2/(1 - pq)$.

Remark: The problem of “gambler’s ruin” is one of the early challenges that led to the theory of probability. In the full statement of the problem, there are $N + 1$ states labeled $0, 1, \dots, N$, with the states 0 and N being absorbing states. At each step there is a probability p of adding one and a probability $q = p - 1$ of subtracting one:



The relevant matrices are

$$R = \begin{pmatrix} q & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 0 & p \end{pmatrix} \quad \text{and} \quad Q = \begin{pmatrix} 0 & p & & & \\ q & 0 & p & & \\ & q & 0 & \ddots & \\ & & \ddots & \ddots & p \\ & & & q & 0 \end{pmatrix},$$

where the blank entries of Q are all zero. For $p \neq q$, one can show that

$$(I - Q)^{-1}R = \frac{1}{p^N - q^N} \begin{pmatrix} q(p^{N-1} - q^{N-1}) & p^{N-1}(p - q) \\ q^2(p^{N-2} - q^{N-2}) & p^{N-2}(p^2 - q^2) \\ \vdots & \vdots \\ q^{N-2}(p^2 - q^2) & p^2(p^{N-2} - q^{N-2}) \\ q^{N-1}(p - q) & p(p^{N-1} - q^{N-1}) \end{pmatrix}.$$

In the limit $p = q = 1/2$, this becomes

$$(I - Q)^{-1}R = \begin{pmatrix} (N-1)/N & 1/N \\ (N-2)/N & 2/N \\ \vdots & \vdots \\ 2/N & (N-2)/N \\ 1/N & (N-1)/N \end{pmatrix}.$$